



NATIONAL UNIVERSITY OF TECHNOLOGY
DEPARTMENT OF INFORMATION TECHNOLOGY
DATASTRUCTURE ALGORITHM
PROJECT REPORT

Project Title:

Hospital Patient Queue Management System

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➤ **Introduction**

Hospitals receive a huge number of patients in a day. There are patients who need an emergency treatment and those who have a normal waiting list. The manual management of patients is more likely to create delays and mistake. Hence, a system that is automated and efficient is needed.

➤ **Problem Statement**

This poses a problem to hospitals that fail to handle patients effectively and in cases where emergency and normal patients are presented at the same time.

Manual systems cause:

- Mismanagement
- Increased waiting time
- None of the prioritizing of the important cases.
- Delays in searching of old patient records.

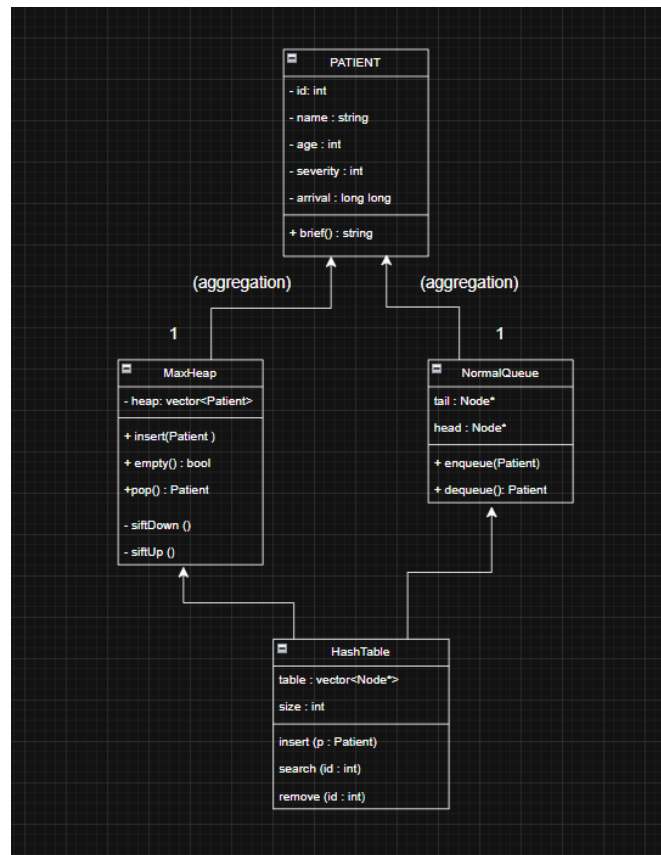
A computerized system is required which can prioritize the emergency patients, handle normal queues, and search patient history very fast with the help of an efficient data structure.

➤ **OBJECTIVE:**

The main objectives of this project are:

- To rank emergency patients using a Max-Heap.
- Handling the regular patients by a FIFO queue.
- Effective storage and retrieval of patient records by using a hash table.
- To find out the waiting time of patients.
- To provide a simple menu based interface to hospital employees.

➤ System workflow:



1. Added new patient (emergency or normal)
2. Serve the next patient according to priority.
3. Keep patient history hash based.
4. Search patient by ID
5. Display waiting patients
6. Estimate waiting time analytics.
7. Release patient on discharge.

➤ **Data Structures Used:**

Data Structure	Purpose	Complexity
Max-Heap (vector)	Emergency patient prioritization	Insert/Extract: $O(\log n)$
Singly Linked List	Normal patient FIFO queue	Enqueue/Dequeue: $O(1)$, Search: $O(n)$
Hash Table	Fast storage & lookup of patient records	Insert/Search/Delete: $O(1)$ avg, $O(n)$ worst
Vector	Store waiting times for analytics	$O(1)$ append, $O(n)$ average computation

➤ **Algorithms Used**

Heap Operations : For emergency patients (insert & extract max).

FIFO Queue Operations : For normal patients (enqueue & dequeue).

Hashing : For fast patient record storage and retrieval using ID.

Linear Search in List / List Removal : For handling collision chains in hash table.

➤ **Expected Output**

The system will provide:

- Proper typing of patients according to severity and arrival.
- Quick and efficient patient history search.
- Reduced waiting times
- Effective record keeping of all treated patients.

➤ **Implementation:**

Language: C++

Key Features:

- Interactive menu so that it is user-friendly.
- Different queues in cases of emergency and normal patients.
- Hash table in order to store and easily retrieve patient records, wait time tracking and analytics report generation
- Validity and safety of inputs

➤ **Conclusion**

The Hospital Management System:

- Max-heap of priority to the emergency patients.
- Holds FIFO normal queue in the case of regular patients.
- Fast record retrieval and discharge
- Uses a hash table. Wait time monitor and analytics generation.